

**NATIONAL PRACTICAL EXAMINATION PREPARATION FOR TECHNICAL
SECONDARY SCHOOLS (TSS)**

SCHOOL YEAR 2024-2025

SUMMATIVE/INTEGRATED ASSESSMENT

SECTOR: ICT AND MULTIMEDIA

TRADE: SOFTWARE DEVELOPMENT

Integrated Assessment Project

Develop a web application using Laravel (PHP framework), MySQL database, Blade templates (or Vue.js), and RESTful APIs.

1. INTEGRATED SITUATION

Smart Parks is a company located in Rubavu District, Western province of Rwanda. It provides multiple car related services.

It is struggling with staff information management, payroll details for employees, and report generation.

The director wants a system developed using Laravel to manage:

- Employees
- Departments
- Salary calculations
- Reports

The system should:

- Calculate salaries based on department
- Store employee data
- Allow editing and deletion
- Provide secure access with authentication

Tasks

As a full stack developer, you are asked to develop a web-based application using Laravel.

1. Design database models using Laravel migrations:

- Employees table
- Departments table

- Salaries table
2. Define relationships using Eloquent ORM:
 - Department hasMany Employees
 - Employee belongs To Department
 3. Create database schema using migrations and run:
`php artisan migrate`
 4. Create Controllers:
 - EmployeeController
 - DepartmentController
 5. Implement CRUD operations for:
 - Employees
 - Departments
 6. Create Blade views for:
 - Forms (input data)
 - Tables (display data)
 7. Use Laravel validation for form inputs.
 8. Implement authentication using Laravel Breeze.
 9. Use MySQL as the database system.
 10. Create RESTful routes:
`Route::resource('employees', EmployeeController::class);`
`Route::resource('departments', DepartmentController::class);`
 11. Implement salary calculation logic inside controllers.
 12. Ensure the application is responsive and user-friendly.
 13. Deploy the system locally using Laravel development server:
`php artisan serve`